

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. When process and sensor noises are correlated **at the same time instant**, what strategy do we take to make unbiased state estimates? (Select all that apply.)

1 / 1 point

☒ We modify Kalman-filter Step 1a with a new $\bar{A}_k$ matrix and an augmented deterministic input.

☒ **Correct**
Yes. This is correct.

☒ We modify Kalman-filter Step 1b to use a new $\Sigma_{\bar{w}}$ matrix instead of $\Sigma_{\bar{w}}$.

☒ **Correct**
Yes. This is correct.

☐ We modify how we compute L_k in Kalman-filter Step 2a.

☐ We compute a different value for $\hat{x}_k^+$ in Kalman-filter Step 2b because of the change in how we compute L_k .

☐ We compute a different value for $\Sigma_{\hat{x},k}^+$ in Kalman-filter Step 2c because of the change in how we compute L_k .

2. When process and sensor noises are correlated **at a time offset of one sample**, what strategy do we take to make unbiased state estimates? (Select all that apply.)

1 / 1 point

☐ We modify Kalman-filter Step 1a with a new $\bar{A}_k$ matrix and an augmented deterministic input.

☐ We modify Kalman-filter Step 1b to use a new $\Sigma_{\bar{w}}$ matrix instead of $\Sigma_{\bar{w}}$.

☒ We modify how we compute L_k in Kalman-filter Step 2a.

☒ **Correct**
Yes. This is correct.

☒ We compute a different value for $\hat{x}_k^+$ in Kalman-filter Step 2b because of the change in how we compute L_k .

☒ **Correct**
Yes. This is correct.

☒ We compute a different value for $\Sigma_{\hat{x},k}^+$ in Kalman-filter Step 2c because of the change in how we compute L_k .

☒ **Correct**
Yes. This is correct.

3. When might we expect process noise and sensor noise to be correlated? (Select all that apply.)

1 / 1 point

☐ When the physical process is affected by changes in temperature and the measurement is affected by inductive electrical interference.

☒ When the physical process and measurement system are both affected by changes in temperature.

☒ **Correct**
Yes. This is one example.

☐ When the physical process is affected by changes in inductive electrical interference and the measurement is affected by changes in temperature.

☒ When the physical process and measurement system are both affected by inductive electrical interference.

☒ **Correct**
Yes. This is one example.